2 .What is a Wallet?

A crypto wallet is essentially software that allows you to connect to the blockchain, manage your assets, and conduct transactions.

Hardware Wallets and Software Wallets

- **Software Wallet :** Connected to the internet, used for daily trading and fast interaction. Software is convenient but requires high security
  Examples: Rabby, MetaMask
  Mobile: Fast and portable. Examples: Base, Trust Wallet, Rainbow, Phantom
- **Hardware Wallet :** Offline, for long-term storage and primary assets Hardware is the most secure option.
  Examples: Ledger, Trezor

❖ **Crucial Tips:**

Never store the private key online.

Use Ledger or a reputable hardware wallet.

Store the Seed Phrase on paper or metal and keep it in a safe place.

The Golden Rule: Always keep only the amount you need in your software wallet; store the rest in your hardware wallet.

3 .Using a VPN

A VPN is not just for secure login to Gmail or X; it protects your digital identity and transactions.

Always use a VPN for account creation, logging into exchanges, and trading.

Utilize a reliable, static IP so algorithms do not flag your account as a risk.

4 .Professional Security Behavior

- Never click on suspicious links.
- Verify emails and messages before replying.
- Research contracts and projects before interacting with them.

❖ **Note:** Security in Web3 is not just technical; it's cultural. It means always being vigilant, even when you feel comfortable.

We no longer talk about Web3 in terms of the previous concepts, because by creating a wallet, you have entered the world of the blockchain. You are on-chain. Your activities are recorded on the blockchain.

Friend, the blockchain might seem very complex at first glance, but if you look at it simply, the blockchain is a very large and transparent ledger that everyone can see, but no one person can tamper with.

The Cornerstone of Trust - Security and Privacy in Digital Wallets

Security and privacy are the fundamental pillars for the widespread adoption of cryptocurrencies. A digital wallet (wallet) is not merely an application; it is the gateway and vault for your digital assets.

1. The Crucial Distinction: Custody vs. Ownership

The most vital point to understand is that your digital assets are stored on the blockchain, not inside the wallet. The wallet is simply the tool used to safeguard your private keys, which serve as proof of ownership over those assets.

Private Key: A string of data that allows you to sign transactions and prove your ownership. This key must remain secret

Public Key: Derived from the private key, it is used to receive funds (like a bank account number).

Seed Phrase/Mnemonic Phrase: A human-readable backup (usually 12 or 24 words) that encrypts all your private keys. Losing or revealing this phrase means the complete loss of your assets.

2. Wallet Security Levels: Hardware vs. Software

Wallet security is generally categorized based on how the private key is stored:

A. Hardware Wallets

This represents the most secure method of storage. The private key never interacts with an internet-connected device

- **Hardware Wallets:** Such as Ledger products, these devices store private keys on a secure chipTransaction signing occurs directly on the device, and the private key never leaves the hardware. This method offers the best balance between security and usability

- **Paper Wallets (Less Recommended):** The private key is printed on paper. The primary issues here are physical risks (fire, water damage) and the inherent difficulty in securely moving the assets later.

B. Software Wallets

These wallets are either continuously or frequently connected to the internet

Software Wallets: Installed on a computer or mobile phone. If the device is compromised by malware or phishing attacks, the private keys are at risk.

Exchange Wallets (Custodial Wallets): The private key is held by a third party (the exchange). This is the most convenient method but violates the principle of Not your keys, not your coins.

3. Privacy Considerations

Privacy in the crypto space often conflicts with the transparency of the blockchain:

Blockchain Transparency: All transactions are permanently recorded on the public ledger, and addresses can be tracked.

Pseudonymity vs. Anonymity: Addresses are not truly anonymous; they are pseudonyms. If an address is linked to your real identity (e.g., through exchanges or websites), your entire transaction history can be exposed.

Solutions: The use of privacy networks like Mixers or Stealth Addresses, and wallets that internally employ technologies like one-time addresses, are essential for enhancing privacy.

4. Security Standards Emphasized (e.g., by Ledger)

Experts like Guillemet typically insist on adhering to these points:

- **Secure Element (SE):** In hardware wallets, the use of a certified, secure chip (similar to EMV in credit cards) to protect the private key against physical and software attacks.
- **Physical Transaction Confirmation:** For every transaction, the user must verify it directly on the physical device screen, not just via a potentially compromised computer display.
- **Strong Passcode (PIN):** Use of a complex PIN for initial device access.
- **Secure Backup:** Storing the seed phrase in a safe, offline, physical location (e.g., in a safe or vault).

❖ **Conclusion:** Your digital security depends on your weakest link. In the realm of wallets, this weak link is almost always where the private key is compromised. A hardware wallet, combined with meticulous management of the seed phrase, represents the best defense against most cyber threats.

Tokens and Layers

Layer 1 (Main Layer)

The base chains like Bitcoin, Ethereum, Solana

Security and consensus happen at this layer

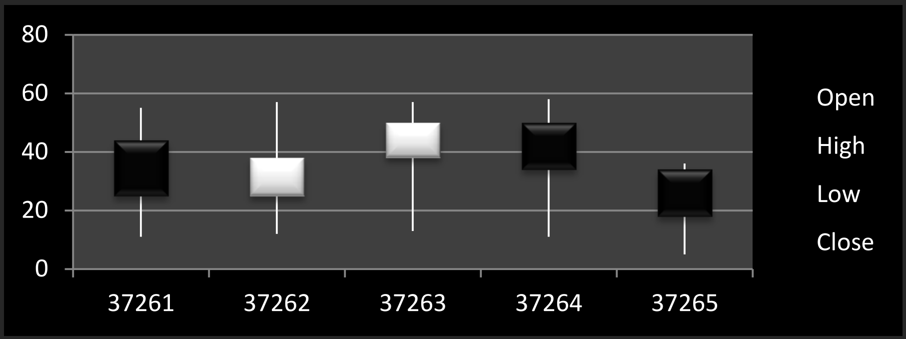
Layer 2 (Development Layer)

Chains built on top of Layer 1 to reduce costs and increase speed

Examples: Base and Arbitrum

Token Types

- Native Token: Such as ETH, BTC, SOL
- ERC-20: Fungible tokens on Ethereum
- ERC-721: Non-fungible tokens (NFTs)
- BRC-20: Bitcoin standard tokens
- SPL: Token standard on Solana



Bitcoin ($BTC)

Digital Gold

The first blockchain in the world (2009) created by Satoshi Nakamoto

- **Goal: A censorship-resistant, intermediary-free payment system**
- **Consensus Model: Proof of Work (PoW) mining via computational power**
- **Programming Language: Limited, lacking advanced smart contracts**

A secure and immutable network, but not designed for complex applications

Native Token: BTC

Use: Store of value, payment, and the base for secondary layer DeFi

(like Bitcoin L2 – Stacks, Rootstock)

Ethereum ($ETH)

Created by Vitalik Buterin in 2015

- Goal: To build a network for running Decentralized Applications (DApps)
- Consensus Model: Proof of Stake (PoS
- Programming Language: Solidity
- Key Feature: Smart Contracts

A Smart Contract is a self-executing contract on the blockchain. Its code is written to execute the action automatically when its conditions are met, without the need for an intermediary

Example: Imagine making a smart contract that states: "If I transfer the money today at 5 PM, automatically send me the design file." No bank or intermediary is needed; everything is executed by code and transparency

Native Token: ETH

Use: Paying gas fees, collateral in DeFi, and store of value

Ethereum-based Networks and Layers

To reduce fees and increase speed, many projects are built on Ethereum

Layer 2 Solutions

- **Arbitrum**: Fast and cheap, for DeFi and NFT
- **Optimism:** For light and scalable smart contracts
- **Base:** Created by Coinbase with simple UX and an active community
- **zkSync:** Using ZK-Rollup technology for privacy and speed

All these networks rely on the security of the main Ethereum layer, but they process transactions off-chain to be faster and cheaper

Solana ($SOL)

High Speed and Scale

Founded in 2017 by Anatoly Yakovenko

- Goal: High transaction speed and very low fees
- Hybrid Consensus: Proof of History (PoH) + Proof of Stake (PoS)
- Programming Language: Rust and C

Designed for high-speed DeFi, gaming, and NFT

- **Native Token: SOL**
- Use: Paying gas, governance, and staking
- Popular Solana Wallets: Phantom, Solflare, Backpack

Simple Example: Traditional Bank

Imagine you have an account at Bank . When you transfer 1 million Tomans to a friend's account, what happens?

- Bank records the transaction
- The bank itself decides if it should proceed or not
- Only the bank has access to the data

If its system fails or makes a mistake, the transaction might be deleted or reversed

Conclusion: You must trust the bank because it holds the ledger. This is a Centralized system

What happens in the Blockchain World ?

The blockchain comes in to say: "We no longer trust one specific person or organization to hold the ledger." Thousands or millions of computers worldwide hold a copy of that ledger, and every transaction you make must be validated among them